

Problem Statement 8: Detection and Characterization of Subsurface Ice

This document compiles the research papers relevant to Problem Statement 8, categorized by their contribution to the specific objectives of the project.

1. The Primary Source Paper

Sinha, Bharti, Acharyya, Mishra, Srivastava & Bhardwaj (2026), npj Space Exploration

Subsurface ice in doubly shadowed craters as revealed by Chandrayaan-2 dual frequency synthetic aperture radar

- **Description:** This very recent paper examines nine "doubly shadowed craters" within Faustini, Haworth, and Shoemaker craters. It establishes a refined diagnostic criterion for volumetric scattering indicative of subsurface ice by pairing Circular Polarization Ratio ($CPR > 1$) with Degree of Polarization ($DOP < 0.13$).
 - **Contribution to Project:** This is the foundational paper upon which the hackathon problem statement is built. It provides the exact threshold criteria ($CPR > 1$ and $DOP < 0.13$) and the methodological framework needed for Objective 1 and Objective 2. Replicating their pipeline on the assigned crater is the recommended primary step.
 - **Links/Source:** [Nature - npj Space Exploration](#)
-

2. Objective 1 & 2: Radar-Ice Detection and Ambiguity

Spudis et al. (2013), JGR Planets

- **Description:** A foundational Mini-RF (LRO) paper that distinguishes two classes of high-CPR craters: fresh craters with rocky ejecta (high CPR inside and outside) and craters with high CPR exclusively in permanently dark, cold interiors (consistent with water ice).
- **Contribution to Project:** Provides a critical sanity check to ensure the crater's elevated CPR is contained within the rim, supporting the ice hypothesis over rocky terrain.
- **Links/Source:** [Wiley Online Library](#)

Fa & Eke (2018), JGR Planets

- **Description:** Argues that elevated CPR in young craters can sometimes reflect an intermediate stage of crater degradation rather than the presence of ice.
- **Contribution to Project:** A key reference to understand the ambiguity of CPR alone and why pairing it with DOP is necessary.

Thomson et al. (2012)

- **Description:** A Shackleton crater case study showing that CPR on crater walls differs from the floor, ruling out surface ice there.
- **Contribution to Project:** Reinforces the need for spatial discrimination within the crater rather than relying on a single bulk CPR value.

Mishra, Kumar & Singh (2014), IEEE JSTARS

- **Description:** Proposed an earlier ice indicator using $CPR > 1$ combined with $DOP < 0.35$.
 - **Contribution to Project:** Demonstrates the evolution of the diagnostic threshold, explaining why the 2026 source paper tightened the DOP requirement to < 0.13 .
-

3. Objective 1: Mapping "Doubly Shadowed Craters"

O'Brien & Byrne (2022), Planetary Science Journal

- **Description:** Applied illumination models to high-resolution LOLA topography to produce the first map of double shadows at the lunar poles.
- **Contribution to Project:** Provides the blueprint methodology (horizon-profile/gnomonic-projection illumination modeling) for mapping PSRs and identifying doubly shadowed craters using DEM and OHRC data.
- **Links/Source:** [IOPscience](#)

Carruba & Coradini (1999) & Sefton-Nash et al. (2019)

- **Description:** Discuss theoretical treatment of "double shielding" cold-trap physics and present ultra-cold trap evidence in Amundsen crater.
 - **Contribution to Project:** Offers theoretical backing for why doubly shadowed craters are ideal candidates for long-term volatile preservation.
-

4. DFSAR-Specific Processing Methodologies

Putrevu et al. (2023), JGR Planets & Verma et al. (2025), Icarus

- **Description:** Prior DFSAR surveys of polar craters utilizing full-polarimetric capability (HH, HV, VH, VV) and establishing data processing pipelines (multilooking, radiometric correction, orthorectification).
- **Contribution to Project:** Essential for processing the raw Chandrayaan-2 DFSAR data provided in the hackathon and correctly computing parameters.
- **Links/Source:** [Wiley Online Library](#)

Bhiravarasu et al. (2021), Planetary Science Journal

- **Description:** Details DFSAR instrument performance and calibration.
 - **Contribution to Project:** Useful for understanding resolution and incidence-angle constraints during data processing.
-

5. Objective 3: Landing Site Safety

Chakraborty et al. (2024), Planetary and Space Science

- **Description:** *Polarimetric Characterization of Chandrayaan-3 Landing Site near Lunar South Pole using High Resolution Chandrayaan-2 DFSAR data.*

- **Contribution to Project:** The closest existing template for Objective 3. It demonstrates how to use radar polarimetry as a substitute for optical imagery to assess hazard avoidance (smooth, rock-free terrain) in PSRs, where sunlight is absent.
- **Links/Source:** [ScienceDirect](#)

Rajasekhar et al. (2024), *Icarus*

- **Description:** Companion paper analyzing the actual Chandrayaan-3 site for morphology, hydration, and gravity anomalies.
- **Contribution to Project:** Serves as a methodological template for multi-parameter landing site evaluation.

6. Objective 4: Rover Traverse Planning

Lamarre, Malhotra et al. (2023–2024), *arXiv/AERO/IEEE*

- **Description:** Addresses solar-powered rover path planning into PSRs (within the VIPER mission context), highlighting the need for global spatiotemporal planning to ensure continuous solar exposure.
- **Contribution to Project:** Provides an illustrative framework for balancing safety vs. efficiency when planning the rover's path into the target doubly shadowed crater.
- **Links/Source:** [arXiv](#)

Companion paper (arXiv 2307.16786)

- **Description:** *Recovery Policies for Safe Exploration...* presents case studies on traverse path design accounting for terrain hazards, illumination, and communication visibility.
- **Contribution to Project:** Offers a scaling reference for path length and hazard mitigation strategies.
- **Links/Source:** [ResearchGate](#)

Frontiers (2026) accessibility/energy-cost paper

- **Description:** Analyzes realistic traverses and energy requirements for high-priority targets.
- **Contribution to Project:** Highly useful for sanity-checking whether the proposed rover traverse is realistic within a solar power budget constraint.
- **Links/Source:** [Frontiers](#)

7. Objective 5: Ice Volume & Dielectric Estimation

Thompson, Ustinov & Heggy (2011), *JGR Planets*

- **Description:** Foundational radar backscatter modeling for icy lunar regoliths at 13 cm and 4 cm wavelengths.
- **Contribution to Project:** Provides the scattering-model basis for converting CPR observations into an estimated ice fraction.

Kumar et al. (2021), *IEEE TGRS* & Sharma et al. (2022), *Icarus*

- **Description:** Papers focused on estimating dielectric constants using Chandrayaan-2 SAR Data and Integral Equation Modelling.
- **Contribution to Project:** Directly relevant to converting radar returns into a dielectric constant, which is the necessary intermediate step before calculating the subsurface ice volume in the top 5 meters.